MOUNT ZION COLLEGE OF ENGINEERING & TECHNOLOGY

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AI CAREER ROADMAP PLANNER

ABSTRACT

The AI Career Roadmap Planner is a web application that helps students plan their careers using Artificial Intelligence. It is developed using ***HTML, CSS, JavaScript, Python Flask,*** and the ***Claude AI API***. Users can select a target career, enter their current skills and interests, and choose a learning duration. The application generates a personalized roadmap with learning activities, required skills, and practical projects. It helps students identify skill gaps and prepare for future career opportunities. The application also allows users to download their career roadmaps for future reference.

PROBLEM STATEMENT

Students often face difficulties in choosing the right career path and identifying the skills required to achieve their career goals. ***Traditional career guidance may not provide personalized learning plans*** based on individual skills, interests, and learning duration. Therefore, an AI-powered Career Roadmap Planner is developed to generate personalized career roadmaps, recommend relevant skills and practical projects, and help students prepare effectively for their desired careers.

PROPOSED SOLUTION

The AI Career Roadmap Planner is a web-based application that uses Claude AI to generate personalized career roadmaps based on users' career goals, current skills, interests, and learning duration. It provides structured learning plans, recommended technologies, practical projects, and career preparation guidance. Developed using HTML, CSS, JavaScript, and Python Flask, the application helps students ***identify skill gaps, organize their learning, and prepare for their desired careers***. Users can also download their roadmaps for future reference.

KEY FEATURES

- **Personalized Career Planning**: Generates career roadmaps based on individual goals and interests.

- **Skill Gap Identification**: Helps users identify the skills needed for their desired careers.
- **AI-Powered Roadmap Generation**: Uses Claude AI to create structured learning plans.
- **Flexible Learning Duration**: Allows users to choose their preferred learning period.
- **Project Recommendations**: Suggests practical projects and relevant technologies.
- **Roadmap Download**: Allows users to download their personalized career plans for future reference.

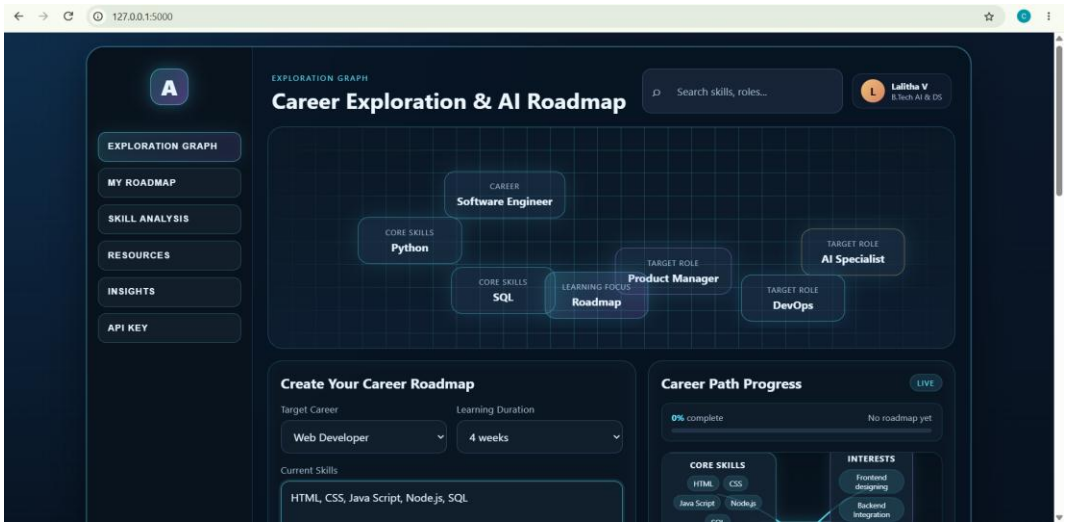
WORKFLOW

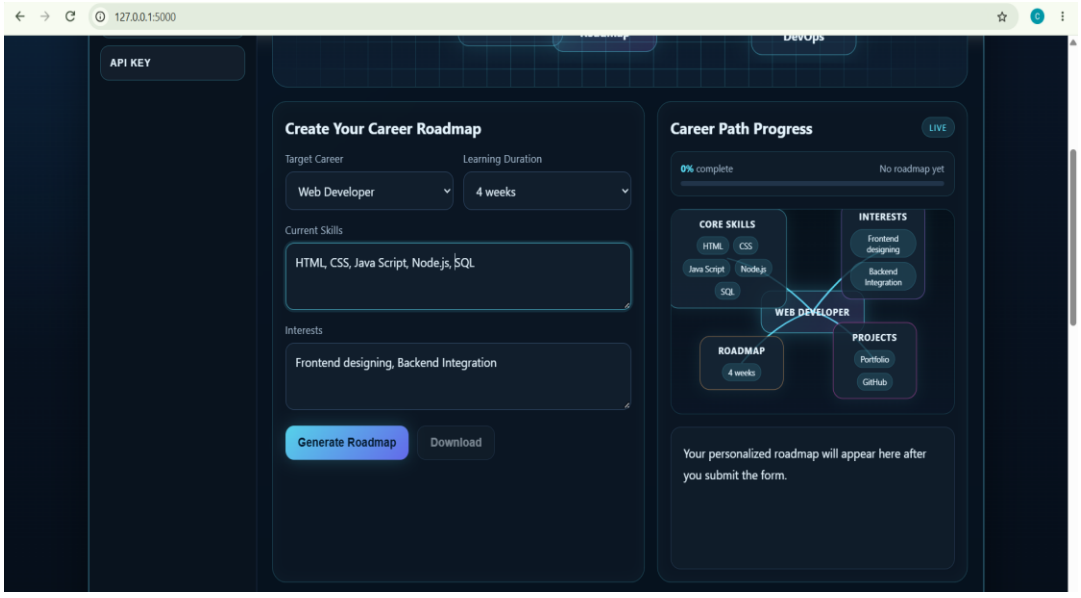


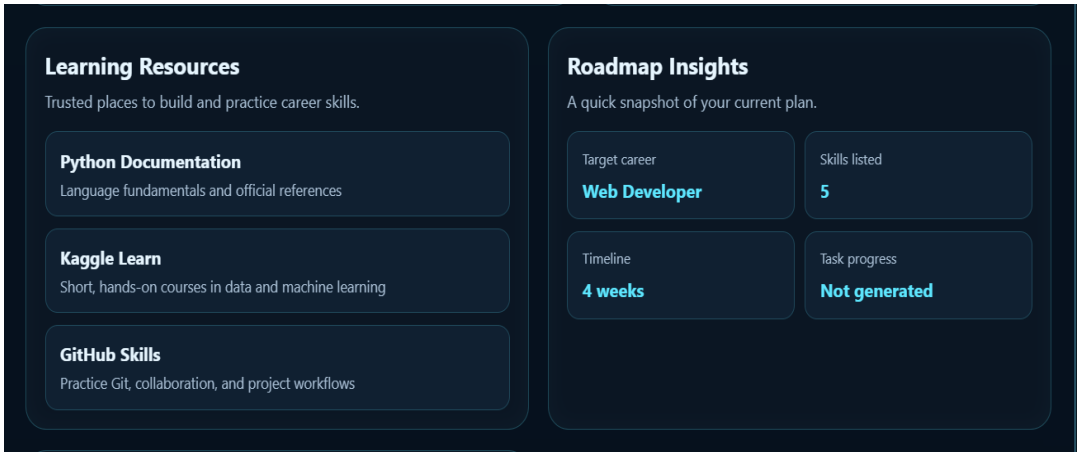
FRONTEND

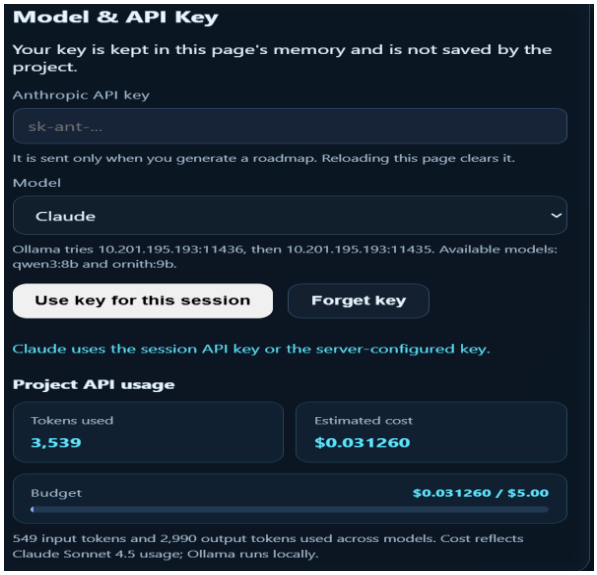
The frontend of the AI Career Roadmap Planner is developed using HTML and CSS to create a user-friendly, attractive, and responsive interface. HTML structures the webpage, including the career selection form, skills and interests fields, learning duration options, and roadmap display section. CSS enhances the appearance through suitable colors, layouts, spacing, and styling to provide a better user experience.

JavaScript adds interactivity to the application by collecting user inputs and sending them to the Python Flask backend. It displays the AI-generated career roadmap, provides status and error messages, and enables users to download their personalized roadmaps as text files. The frontend allows users to interact easily with the application and access career guidance based on their individual goals.







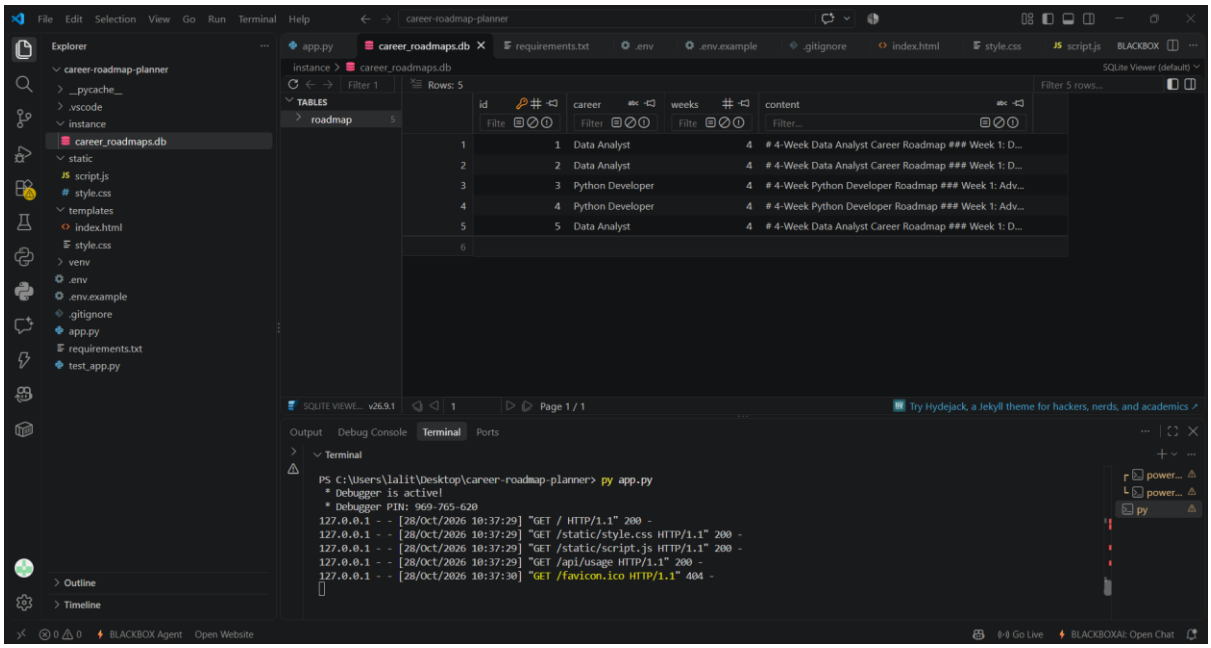


CLAUDE API

The Claude API, developed by Anthropic, is integrated into the AI Career Roadmap Planner to provide intelligent and personalized career guidance. It analyzes users' target careers, existing skills, interests, and preferred learning duration to generate customized learning roadmaps.

The API connects to the Python Flask backend using the Anthropic Python SDK. It generates structured learning plans, recommends relevant technologies and practical projects, and provides career preparation guidance. The API key is securely stored in the .env file to protect sensitive credentials and enable secure communication with the Claude AI service.

DATABASE



The AI Career Roadmap Planner is a web-based application that generates personalized career roadmaps based on users' career goals, existing skills, interests, and preferred learning duration. It uses HTML, CSS, and JavaScript to provide an interactive and user-friendly frontend. The backend is developed using Python Flask, which connects the application with Claude AI to generate customized learning plans. ***SQLite is used to store generated roadmaps, including career names, learning duration, and roadmap content.*** The application helps students organize their learning journey and download their career plans for future reference.

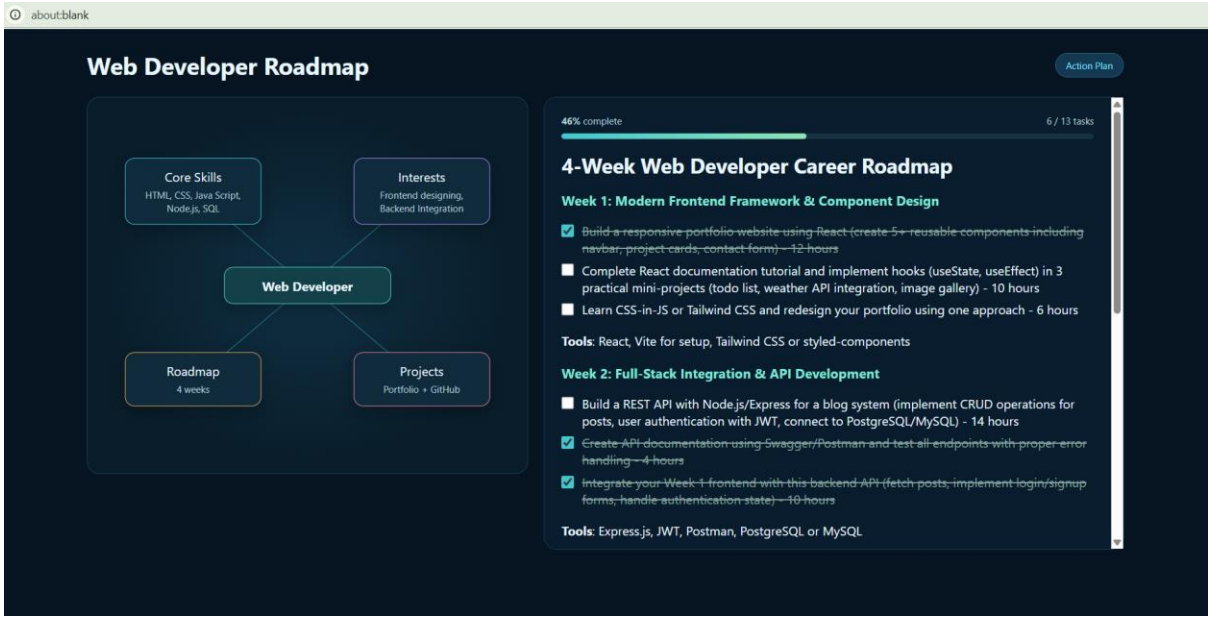
BACKEND

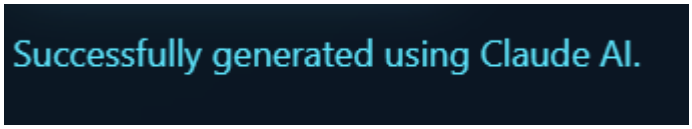
The backend of the AI Career Roadmap Planner is developed using Python Flask, which manages server-side operations and handles requests from the frontend. It receives user inputs, including target career, current skills, interests, and learning duration, and processes them to generate a personalized career roadmap. Flask also connects the application with the Claude AI API to provide intelligent career guidance.

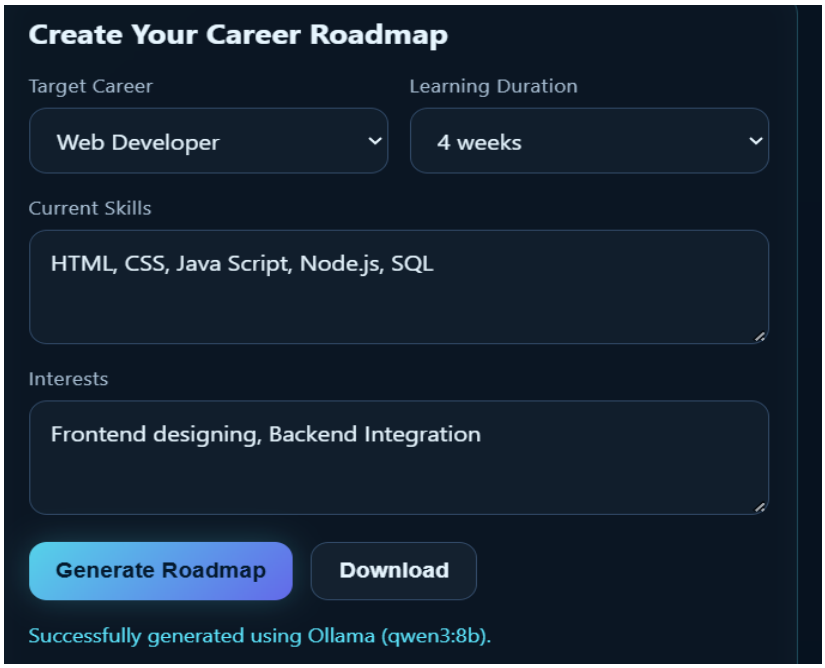
SQLite is used as the database to store generated career roadmaps, including career names, learning durations, and roadmap content. The backend retrieves and stores the required information and sends the generated results to the frontend in JSON format. It

also handles errors and supports roadmap generation and downloading, ensuring smooth communication between the user interface, AI service, and database.

OUTCOME









CONCLUSION

The AI Career Roadmap Planner provides an effective and personalized approach to career planning using Artificial Intelligence. By integrating HTML, CSS, JavaScript, Python Flask, Claude AI API, and SQLite, the application generates career roadmaps based on users' skills, interests, and goals. It helps students identify required skills, organize their learning activities, and prepare for future career opportunities. Overall, this project demonstrates how AI technology can simplify career planning and support students in achieving their professional goals.